UIT2521 – Information Theory and Applications

UNIT II INFORMATION THEORY FUNDAMENTALS

Tutorial – I

Self-Information

Date & Time: 14.07.2025 (Monday, 3rd Hour)

1. Consider a binary system emitting two symbols **{0, 1}** with probabilities **0.6** and **0.4** respectively. Find the information conveyed by bit '0' and bit '1'.

2. Consider a source emitting two symbols $\mathbf{s_0}$ and $\mathbf{s_1}$ with the corresponding probabilities **¾** and **¼** respectively. Find the self-information of the symbols in

   a) Bits
   b) Decits
   c) Nats

3. A pair of dice is rolled simultaneously. The outcome of the first dice is considered to be '**x**' and that of the second as '**y**'. Three events are defined as

   P = {(x, y) such that (x + y) is exactly divisible by 3}
   Q = {(x, y) such that (x + y) is an even number}
   R = {(x, y) such that (x + y) = 7}

   Which event conveys more information? Justify your answer with mathematical calculation.

1. Information Conveyed by bits '0' and '1'

Given probabilities:

- $P(0) = 0.6$
- $P(1) = 0.4$

The **self-information** ($I(x)$) for symbol x with probability $p(x)$ is:

$$I(x) = -\log_2 p(x)$$

- For bit '0': $I(0) = -\log_2(0.6) \approx 0.74$ bits
- For bit '1': $I(1) = -\log_2(0.4) \approx 1.32$ bits

Interpretation:
Bit '1' conveys more information because it is less probable than bit '0'.

2. Self-Information of Symbols s_0 and s_1

Given probabilities:

- $P(s_0) = 3/4 = 0.75$
- $P(s_1) = 1/4 = 0.25$

a) Bits (log base 2)

- $I(s_0) = -\log_2(0.75) \approx 0.42$ bits
- $I(s_1) = -\log_2(0.25) = 2.00$ bits

b) Decits (log base 10)

- $I(s_0) = -\log_{10}(0.75) \approx 0.12$ decits
- $I(s_1) = -\log_{10}(0.25) \approx 0.60$ decits

c) Nats (log base e)

- $I(s_0) = -\ln(0.75) \approx 0.29$ nats
- $I(s_1) = -\ln(0.25) \approx 1.39$ nats

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3. Dice Events and Information Conveyed

- **Total outcomes:** 36 (since each die has 6 sides)
- **Events:**

Event	Criteria	Favorable Outcomes	Probability	Self-Information ($-\log_2(P)$)
P	$x + y$ divisible by 3	12	0.33	1.58 bits
Q	$x + y$ even	18	0.5	1.00 bit
R	$x + y = 7$	6	0.17	2.58 bits

- **P:** 12 outcomes ($P = 12/36 = 1/3$). Information ≈ 1.58 bits
- **Q:** 18 outcomes ($Q = 18/36 = 1/2$). Information = 1.00 bit
- **R:** 6 outcomes ($R = 6/36 = 1/6$). Information ≈ 2.58 bits

Conclusion:

- **Event R ($x + y = 7$) conveys the most information** because it is the most unlikely event among the three.

Summary Table

Part	Quantity	$s_0/0$ Value	$s_1/1$ Value	Units
1	Self-Information	0.74	1.32	bits
2a	Self-Information	0.42	2.00	bits
2b	Self-Information	0.12	0.60	decits
2c	Self-Information	0.29	1.39	nats
3	Self-Information (P,Q,R)	1.58 (P), 1.00 (Q), 2.58 (R)	–	bits

Thus, event R is most informative mathematically.